

Apolipoprotein(a) Kinetics in Statin-Treated Patients With Elevated Plasma Lipoprotein(a) Concentration

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Background: Lipoprotein(a) [Lp(a)] is a low-density lipoprotein–like particle containing apolipoprotein(a) [apo(a)]. Patients with elevated Lp(a), even when treated with statins, are at increased risk of cardiovascular disease. We investigated the kinetic basis for elevated Lp(a) in these patients.

Objectives: Apo(a) production rate (PR) and fractional catabolic rate (FCR) were compared between statin-treated patients with and without elevated Lp(a).

Methods: The kinetics of apo(a) were investigated in 14 patients with elevated Lp(a) and 15 patients with normal Lp(a) levels matched for age, sex, and body mass index using stable isotope techniques and compartmental modeling. All 29 patients were on background statin treatment. Plasma apo(a) concentration was measured using liquid chromatography–mass spectrometry.

Results: The plasma concentration and PR of apo(a) were significantly higher in patients with elevated Lp(a) than in patients with normal Lp(a) concentration (all $P < 0.01$). The FCR of apo(a) was not significantly different between the groups. In univariate analysis, plasma concentration of apo(a) was significantly associated with apo(a) PR in both patient groups ($r = 0.699$ and $r = 0.949$, respectively; all $P < 0.01$). There was no significant association between plasma apo(a) concentration and FCR in either of the groups ($r = 0.160$ and $r = -0.137$, respectively).

Conclusion: Elevated plasma Lp(a) concentration is a consequence of increased hepatic production of Lp(a) particles in these patients. Our findings provide a kinetic rationale for the use of therapies that target the synthesis of apo(a) and production of Lp(a) particles in patients with elevated Lp(a). (*J Clin Endocrinol Metab* 104: 6247–6255, 2019)

Lipoprotein(a) [Lp(a)] is a highly polymorphic particle consisting of a low-density lipoprotein (LDL) particle containing apolipoprotein(a) [apo(a)] covalently linked to apoB-100 (1, 2). Lp(a) possesses atherogenic, thrombotic, and proinflammatory properties (2, 3). Classic and Mendelian epidemiological studies suggest that elevated Lp(a) concentration is causally related to atherosclerotic

cardiovascular disease (ASCVD) and calcific aortic stenosis (3–6). Recent large clinical trials have consistently shown that elevated Lp(a) concentration remains a risk factor despite reduction of LDL-cholesterol with statins (7, 8).

Plasma Lp(a) concentrations are highly heritable, being chiefly determined by variations in the number of

plasminogen kringle IV–related (KIV) type 2 repeats in the *LPA* gene (1, 2). Accordingly, there is a major inverse association between the number of KIV type 2 repeats in apo(a) and plasma Lp(a) concentrations (5, 6, 9). Although Lp(a) and LDL share similarities in structure, the role of the LDL receptor in Lp(a) particle catabolism remains unclear (10–13). Statins enhance the expression and activity of LDL receptors, but they do not significantly decrease plasma Lp(a) concentration (7), with some reports suggesting that they may increase Lp(a) concentration (14). The exact mechanism responsible for elevated Lp(a) levels in humans is poorly understood (2, 15), particularly in patients receiving statin therapy. Understanding the metabolism of Lp(a) in humans requires investigation of Lp(a) particle kinetics. There is a scarcity of studies on Lp(a) kinetics using stable isotope tracers and compartmental modeling methods (2, 16). A recent recommendation by an expert group identified several knowledge gaps in Lp(a) research, underscoring the importance of investigating the mechanistic basis for elevated Lp(a) concentration in patients treated with statins as a rationale for the use of new therapies (17).

To address the aforementioned knowledge gaps, we used a stable isotope technique to test the hypothesis that the hepatic production of apo(a) is increased in patients with elevated Lp(a) concentrations who are taking statins. We also investigated the association of apo(a) isoforms and Lp(a) kinetics.

Methods

Patients

Fifteen white patients (8 men and 7 women) aged between 18 and 70 years were recruited from the Lipid Disorders Clinic at Royal Perth Hospital. All patients had elevated plasma Lp(a) concentrations of ≥ 0.8 g/L (or ≥ 170 nmol/L; *i.e.*, 95th population percentile), were taking statins and aspirin, and had moderate to high ASCVD risk or history of ASCVD events. One patient was withdrawn during the study because of unexpected illness not related to the study. Fifteen statin-treated patients matched for age (to within 5 years), sex (8 men and 7 women), body mass index (to within 3 kg/m²), and total and LDL-cholesterol (to within 0.5 mmol/L) levels with normal Lp(a) concentrations of ≤ 0.3 g/L were also recruited from the lipid clinic. No patients had diabetes, renal disease (plasma creatinine >130 μ mol/L), hepatic dysfunction (alanine aminotransferase $>3\times$ the upper limit of normal), anemia (hemoglobin <125 g/L), hematological disorders, or cardiovascular disease events in the past 6 months; nor were they current smokers or taking other medications affecting Lp(a) metabolism (*e.g.*, hormonal replacement therapy, niacin, proprotein convertase subtilisin/kexin type 9, and monoclonal antibodies). All patients provided written informed consent, and the study was approved by the Royal Perth Hospital Human Research Ethics Committee.

Clinical protocol

Eligible patients were admitted to the site metabolic ward after a 14-hour fast. They were studied in a semirecumbent position and allowed to drink only water. Venous blood was collected for laboratory measurements. A single bolus of ²H₃-leucine (5 mg/kg of body weight) was administered IV within a 2-minute period into an antecubital vein (18, 19). Blood samples were taken at baseline and at 5, 10, 20, 30, and 40 minutes and at 1, 1.5, 2, 2.5, 3, 4, 5, 6, 8, and 10 hours after injection of the isotope. Patients were then given a meal and discharged home. Additional fasting blood samples were collected in the morning on the following 4 days of the same week (*i.e.*, at 24, 48, 72, and 96 hours after injection of the isotope).

Isolation and measurement of isotopic enrichment of Lp(a)

Lp(a) was isolated by an immunoprecipitation method, as previously described (18, 19). Briefly, 250 μ L of human Lp(a) antibody (Advy Chemical Ltd, Mumbai, India) were immobilized onto 80 μ L of magnetic beads G proteins (MBGPs) by incubation for 1.5 hours at room temperature. The magnetic beads coated with Lp(a) antibody-MBGP were then incubated overnight with 80 μ L of plasma at 4°C, thereby separating Lp(a) from plasma as an Lp(a)-Lp(a) antibody-MBGP complex. Lp(a) was eluted from the beads and then reduced by dithiothreitol in sample buffer. The apo(a) protein was separated by sodium dodecyl sulfate gel electrophoresis and transferred onto polyvinylidene fluoride membrane by Western blotting. The apo(a) band was excised and hydrolyzed with 200 μ L of 6M HCl at 110°C overnight. After evaporation of HCl, the amino acids were derivatized using an oxazolinone method. Isotopic enrichment was determined as described previously (20).

Biochemical analyses

Plasma total cholesterol, triglyceride, total apoB, high-density lipoprotein (HDL) cholesterol, insulin, and glucose concentrations were determined by standard enzymatic methods (Architect c16000 Analyzer; Abbott Diagnostics, Abbott Laboratories, Abbott Park, IL) and LDL-cholesterol by Friedewald calculation. Plasma Lp(a) concentration was measured as particle number in nanomoles per liter by a direct-binding double monoclonal antibody-based ELISA (Northwest Lipid Metabolism and Diabetes Research Laboratories, University of Washington, Seattle, WA). The capture monoclonal antibody (a-6) was directed to an epitope present in apo(a) K4 type 2, and the detection antibody (a-40) was directed to an epitope present in apo(a) KIV type 9 (21).

Plasma apo(a) was determined by liquid-chromatography-tandem mass spectrometry analyses that utilize a synthetic peptide (LFLEPTQADIALLK) that targets the proteolytic domain of Lp(a) (Proteomics International, Queen Elizabeth II Medical Centre, Perth, WA) (16, 18). Apo(a) isoform size was determined by high-sensitive sodium dodecyl sulfate–agarose gel electrophoresis followed by immunoblotting (Northwest Lipid Metabolism and Diabetes Research Laboratories, University of Washington), and the apo(a) isoforms were expressed in terms of the respective number of KIV repeats as previously reported (22). In three patients, only one apo(a) isoform was expressed. The remaining patients ($n = 26$) had two circulating apo(a)s of different sizes. The predominant apo(a) isoform with the highest density, which contributed substantially to total plasma apo(a)

concentration (>85%), was used for statistical analysis. The coefficient of variation for all biochemical analyses was <5%.

Kinetic modeling

A compartmental model was constructed using SAAM II software (The Epsilon Group, Charlottesville, VA) and fitted to the apo(a) tracer data (16, 18, 19). Two linked models were used to account for the leucine tracer data, including plasma leucine and apo(a) leucine enrichment. The fractional catabolic rate (FCR) of apo(a) was estimated after fitting the model to the apo(a) tracer/tracee ratio data. Models were selected using the Akaike information criterion (23), and goodness of fit was tested by confirming that the residuals were randomly distributed around a mean of zero. The hepatic production rate of apo(a) was calculated as the product of FCR and pool size of apo(a). Pool size was derived by multiplying plasma volume [body weight (kg) $\times$ 0.045] and the plasma apo(a) concentration (nanomoles per liter). A correction factor was applied to adjust for the decrease in relative plasma volume associated with subjects who were overweight/obese (24).

Statistical analysis

Data were analyzed using SPSS 21 software (SPSS, Chicago, IL). Data were presented as mean $\pm$ SD unless otherwise indicated. The Shapiro-Wilk test was used to determine whether the variables were normally distributed. Group characteristics were compared by *t* tests after logarithmic transformation of skewed variables where appropriate. The sample size calculation was based on the hypothesis that patients with elevated Lp(a) concentration had a higher production rate (PR) than patients with normal Lp(a) concentration. Given a PR of apo(a) of 0.83 nmol/kg/d in patients with elevated Lp(a) level and a PR of apo(a) of 0.25 nmol/kg/d in patients with normal Lp(a) level

(19), with a within group SD of 0.20 nmol/kg/d, a sample size of 14 patients per group was estimated to provide >90% power to detect a difference in apo(a) PR, with an α error of 5%. Differences in clinical characteristics between patients with elevated and normal Lp(a) concentrations were tested with χ^2 statistics or independent *t* tests. Associations were examined by simple and partial linear regression methods. Statistical significance was defined at the 5% level using a two-tailed test.

Results

Clinical and biochemical characteristics

Table 1 shows the clinical and biochemical characteristics of the 29 patients (16 men and 13 women) studied. They were, on average, 55 years old, overweight/obese, normotensive, and nondiabetic and had normal renal function. Of these, 15 patients were classified as having probable/definite familial hypercholesterolemia (FH) according to the Dutch Lipid Clinic Network phenotypic classification (25). The remaining patients had mixed hyperlipidemia (*i.e.*, familial combined hyperlipidemia and metabolic syndrome; *n* = 7) and common (or polygenic) hypercholesterolemia (*n* = 7). All patients were receiving statins (atorvastatin, *n* = 5; simvastatin, *n* = 2; rosuvastatin, *n* = 22) and aspirin, with 66% taking ezetimibe. Patients with elevated Lp(a) levels had smaller predominant apo(a) isoform size than patients with normal Lp(a) levels. There were no significant differences in other clinical and biochemical characteristics, including lipid and lipoprotein concentrations, use of medications, and probability of having FH.

Table 1. Clinical and Biochemical Characteristics of the Patients

	Elevated Lp(a) (<i>n</i> = 14)	Normal Lp(a) (<i>n</i> = 15)	<i>P</i> Value
Age, y	56 $\pm$ 11	55 $\pm$ 12	0.833
Male, %	57	53	0.362
Body mass index, kg/m ²	30 $\pm$ 6.4	28 $\pm$ 5.0	0.389
Systolic blood pressure, mm Hg	132 $\pm$ 15	128 $\pm$ 11	0.414
Diastolic blood pressure, mm Hg	75 $\pm$ 8.4	73 $\pm$ 5.8	0.523
Glucose, mmol/L	5.5 $\pm$ 0.4	5.5 $\pm$ 0.4	0.700
Insulin, mU/L	8.2 $\pm$ 4.4	8.8 $\pm$ 4.8	0.744
HOMA score	2.0 $\pm$ 1.1	2.2 $\pm$ 1.4	0.743
Serum creatinine, μ mol/L	74 $\pm$ 19	78 $\pm$ 14	0.466
DLCN score ^a	6.4 $\pm$ 3.5	6.3 $\pm$ 4.7	0.988
Probable/definite FH, % ^a	50	53	0.856
Ezetimibe, %	62	67	0.893
Total cholesterol, mmol/L	4.4 $\pm$ 1.0	4.4 $\pm$ 1.3	0.959
Triglycerides, mmol/L	1.4 $\pm$ 0.8	1.2 $\pm$ 0.5	0.286
HDL-cholesterol, mmol/L	1.2 $\pm$ 0.2	1.3 $\pm$ 0.4	0.311
LDL-cholesterol, mmol/L	2.5 $\pm$ 0.8	2.5 $\pm$ 0.9	0.991
Total apoB, g/L	0.9 $\pm$ 0.2	0.8 $\pm$ 0.3	0.754
Lipoprotein(a), nmol/L ^b	235 (184, 299)	13 (8.0, 22)	<0.001
Predominant apo(a) isoform (KIV repeats)	17 (16, 18)	23 (20, 26)	<0.001

Data are presented as mean $\pm$ SD or geometric mean (95% CI).

Abbreviations: DLCN, Dutch Lipid Clinic Network; HOMA, homeostatic model assessment.

^aRefers to the DLCN score and classification without the genetic analysis component.

^bLp(a) concentration was measured by a direct-binding double monoclonal antibody-based ELISA (21).

Kinetic parameters of apo(a)

Figure 1 shows the isotopic enrichment of apo(a) over time (0 to 96 hours) following the IV infusion of $^2\text{H}_3$ -leucine. The two tracer curves were of similar contour and peak enrichment, suggesting similar turnover rates of apo(a) between the two groups. Table 2 shows the plasma apo(a) concentration and corresponding kinetic parameters in the patients studied. Compared with patients with normal Lp(a) concentration, patients with elevated Lp(a) level had significantly higher plasma apo(a) concentration and apo(a) PR ($P < 0.001$ for both). However, the FCR of apo(a) was not significantly different between the two groups ($P = 0.693$). When PRs were calculated using an estimate of the plasma volume, which corrects for hematocrit (26), the apo(a) PR was significantly higher in patients with elevated Lp(a) concentration than in those with normal Lp(a) concentration [3.8 (2.6, 5.6) vs 0.24 (0.14, 0.40) nmol/kg/d; $P < 0.001$].

Association between apo(a) concentration and metabolic parameters

Figure 2 shows the associations between apo(a) concentration and kinetic parameters in patients with elevated and normal Lp(a) concentrations. In univariate regression analysis, plasma apo(a) concentration was significantly and positively associated with the PR of apo(a) in patients with elevated Lp(a) concentrations ($r = 0.699$; $P = 0.005$) (Fig. 2A). A significant association was also found between apo(a) concentration and PR in patients with normal Lp(a) concentration ($r = 0.949$; $P = 0.001$) (Fig. 2C). There were no significant associations between plasma apo(a) concentration and the FCR of apo(a) in the two groups ($r = 0.160$ and $r = -0.137$, respectively; $P > 0.05$ for both) (Fig. 2B and 2D). The PR and FCR of apo(a) were

significantly and positively correlated in patients with elevated Lp(a) concentration ($r = 0.749$; $P < 0.01$) (27). This association was not found in patients with normal Lp(a) concentration ($r = 0.140$; $P = 0.619$) (27). In partial regression analyses, plasma apo(a) concentration remained significantly associated with the PR of apo(a) in both case and control groups ($r = 0.700$ and $r = 0.945$, respectively; $P < 0.01$ for both) after adjustment for apo(a) isoform size. Adding FH phenotype as a dichotomous variable (presence or absence of probable/definite FH) in the regression model did not alter the results (data not shown). In a pooled analysis using data from the two groups ($n = 29$), plasma apo(a) concentration was significantly and positively associated with the PR of apo(a) ($r = 0.972$; $P < 0.001$) (Fig. 2E). However, there was no significant association between plasma apo(a) concentration and FCR in the pooled group ($r = -0.079$; $P = 0.682$) (Fig. 2F). In partial regression analyses, plasma apo(a) concentration remained significantly associated with apo(a) PR independent of apo(a) isoform size ($r = 0.943$; $P < 0.001$). This analysis confirmed our findings on the association between plasma concentration of apo(a) and corresponding kinetic parameters in elevated and normal Lp(a) concentration groups (Fig. 2A–2D).

Apo(a) isoform size was significantly and inversely associated with the concentration and PR of apo(a) ($r = -0.741$ and $r = -0.708$, respectively; $P < 0.0001$ for both) (27), but not with the FCR of apo(a) ($r = 0.079$; $P = 0.682$) (27) in a pooled analysis from the two groups. Similar observations were observed between apo(a) isoform size with the concentration ($r = -0.633$; $P < 0.001$), PR ($r = -0.666$; $P < 0.001$), and FCR ($r = -0.097$; $P = 0.676$) of apo(a) in patients with smaller apo(a) isoform size (KIV repeats < 23 ; $n = 21$). However, the correlations of apo(a) isoform size with the concentration ($r = -0.487$), PR ($r = -0.247$), and FCR ($r = 0.508$) of apo(a) did not reach statistical significance in patients with larger apo(a) isoform size, probably owing to small sample size ($n = 8$).

Plasma insulin concentration was not significantly associated with the FCR and PR of apo(a) in patients with elevated Lp(a) (FCR: $r = -0.375$; $P = 0.187$ and PR: $r = -0.200$; $P = 0.492$) or normal Lp(a) concentrations (FCR: $r = 0.184$; $P = 0.513$ and PR: $r = 0.392$; $P = 0.148$). Likewise, no significant associations were observed between plasma insulin and the FCR ($r = -0.114$; $P = 0.556$) and PR ($r = 0.013$; $P = 0.946$) of apo(a) using data pooled from the two groups. Similarly, no statistically significant associations were found between the homeostatic model assessment score and the FCR and PR of apo(a) (data not shown).

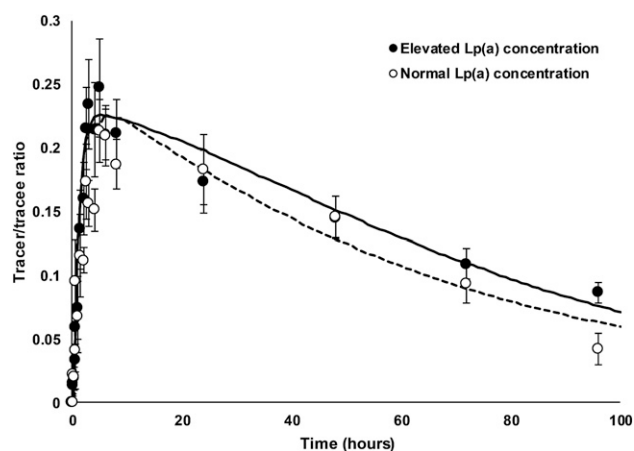


Figure 1. Isotopic enrichment of apo(a) from 0 to 96 hours in all patients studied with elevated (solid line) and normal (dashed line) Lp(a) concentrations. Mean $\pm$ SEM is shown.

Table 2. Plasma Concentration, Production Rate, and Fractional Catabolic Rate of apo(a) in Patients With Elevated and Normal Plasma Lp(a) Concentrations

	Elevated Lp(a) Concentration Group (n = 14)	Normal Lp(a) Concentration Group (n = 15)	P Value
Apo(a) concentration, nmol/L ^a	235 (184, 299)	13 (8.0, 22)	<0.001
Apo(a) production rate, nmol/kg/d	3.8 (2.6, 5.6)	0.24 (0.14, 0.40)	<0.001
Apo(a) fractional catabolic rate, pools per day	0.46 (0.37, 0.56)	0.48 (0.42, 0.56)	0.693

Data are presented as geometric mean (95% CI).

^aApo(a) concentration was measured by a liquid chromatography-mass spectrometry method (18, 19).

Discussion

The major finding of our study was that patients with elevated Lp(a) concentration had higher hepatic production of apo(a) than patients with normal Lp(a) concentration; the catabolism of apo(a) was not significantly different between groups. We also demonstrated that plasma apo(a) concentration was significantly associated with apo(a) production rate, independent of apo(a) isoform size. Furthermore, Lp(a) particle catabolism was not a significant determinant of plasma Lp(a) concentration particles in these patients. Our results fill an important knowledge gap in relation to understanding the potential of new therapies for elevated Lp(a) in high-risk subjects (17).

Previous findings

Previous studies using radioisotopes examined Lp(a) metabolism in normal subjects and patients with FH and demonstrated that variation in Lp(a) plasma concentration is determined mainly by the production of Lp(a) particles (28, 29). Some, but not all, experimental studies have demonstrated that the LDL receptor is not required for the normal catabolism of Lp(a) (13, 28). Rader *et al.* (28, 29) also found that plasma Lp(a) concentration was inversely associated with apo(a) isoform size and that this involved differences in Lp(a) PR. Using endogenous stable isotope labeling, Frischmann *et al.* (30) showed that patients with smaller apo(a) isoform sizes had higher Lp(a) concentrations as a consequence of increased PR and reduced FCR of apo(a). However, owing to small sample size, the differences in PR and FCR between the two groups did not reach statistical significance. In normolipidemic men with a wide range of Lp(a) concentrations, we previously reported that plasma Lp(a) concentration was predominantly determined by the rate of production of Lp(a) particles, irrespective of apo(a) isoform size (19). None of these studies specifically examined the metabolism of Lp(a) in patients with elevated Lp(a) concentration. In a constant-feeding study of healthy individuals, Jenner *et al.* (31) reported that patients with high Lp(a) levels had increased apo(a) PR and

reduced FCR compared with rates in patients without elevated Lp(a) concentrations. We have extended previous studies by using a larger sample size and examining the kinetics of apo(a) in patients with elevated Lp(a) levels. We studied patients who were taking statins to understand which additional therapy should be added to lower Lp(a), such as apo(a) antisense oligonucleotide and proprotein convertase subtilisin/kexin type 9 monoclonal antibody (18, 32, 33).

Kinetic determinants of Lp(a) concentration

Plasma Lp(a) concentration and isoform size are largely determined by genetic variations in the *LPA* gene (1). The inverse association between apo(a) isoform size and Lp(a) concentration is mediated mainly by the hepatic secretion of apo(a) (28, 31). The size of the apo(a) transcripts is inversely associated with hepatic apo(a) mRNA concentration (34, 35) and by implication apo(a) production. Smaller apo(a) isoforms have a shorter retention time in the endoplasmic reticulum, reflecting a process that involves protein folding, glycosylation, and maturation before secretion (36). A greater intracellular degradation of larger apo(a) isoforms has also been reported *in vitro* (37). Hence, these experiments suggest that smaller apo(a) isoforms are secreted more efficiently into the circulation. Consistent with this, we found that patients with elevated Lp(a) concentrations had smaller apo(a) isoform sizes and higher apo(a) PR than patients with normal Lp(a) concentration. We also confirmed a direct association between the concentration and PR of apo(a) in both elevated and normal Lp(a) concentration groups, reinforcing the notion that plasma concentrations of Lp(a) are determined primarily by the rates of production and not catabolism. Importantly, we found that the significant association between Lp(a) concentration and PR was independent of apo(a) isoform size. This suggests that other factors may also be involved in the regulation of apo(a) production. For example, the availability of apoB-100 for coupling to apo(a) could limit the formation of an Lp(a) particle (38). The significant association between the concentration and PR of apo(a) was

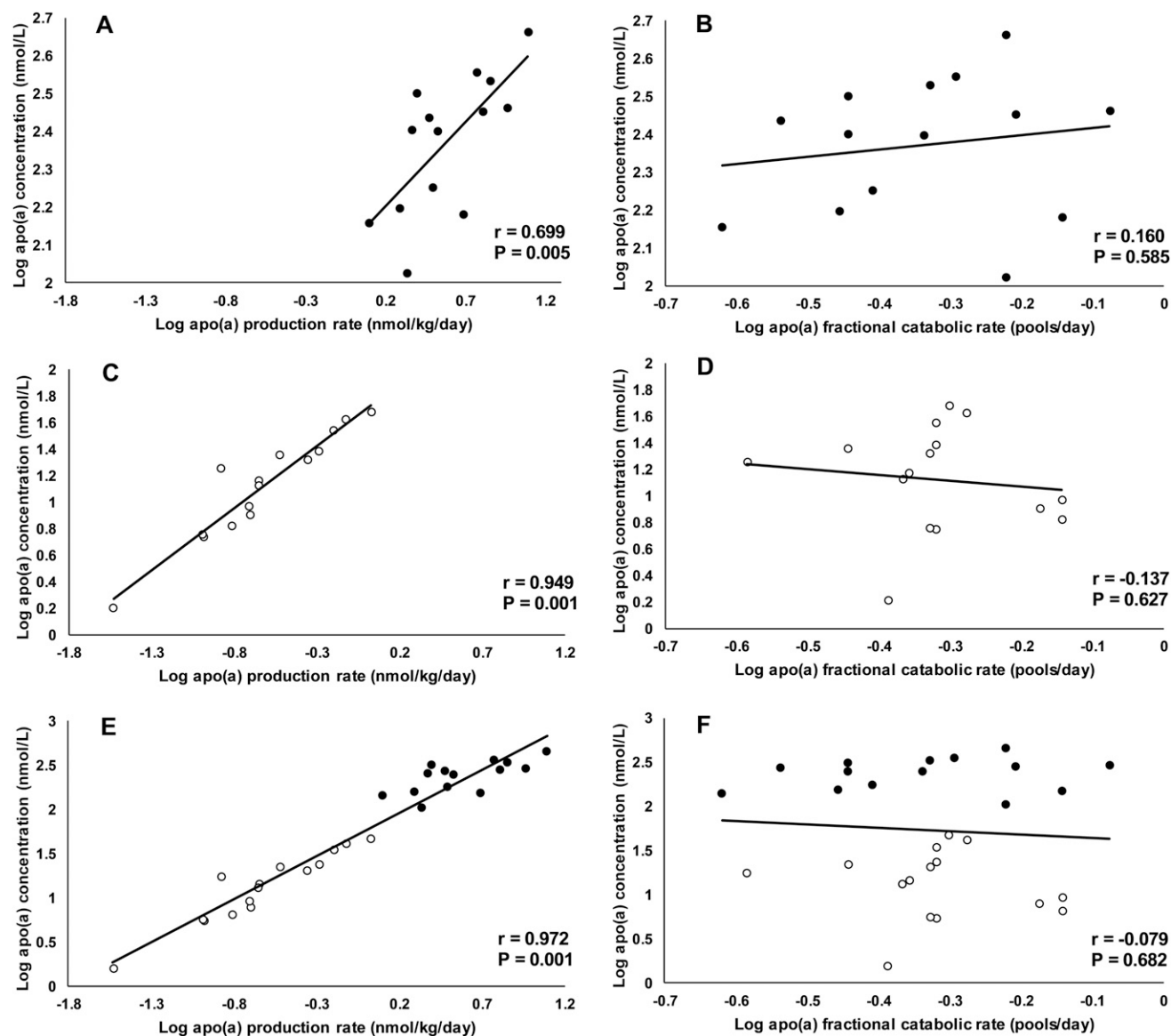


Figure 2. Association of apo(a) concentration and metabolic parameters including production rate and fractional catabolic rate in (A and B) patients with elevated Lp(a) concentrations (black circles), (C and D) patients with normal Lp(a) concentrations (white circles), and (E and F) combined patients.

reconfirmed in the pooled analysis of data from the two groups.

The role of Lp(a) catabolism in determining the concentration of Lp(a) particles remains unclear (2, 13, 15). In the current study, we found that plasma concentration of Lp(a) was not significantly associated with the FCR of apo(a). We also found that the FCR of apo(a) was not significantly different between the two groups. However, our results are not consistent with those of another study showing that the FCR of apo(a) was lower in patients with elevated Lp(a) concentration (31). Discrepancies may relate to different experimental protocols (fasting vs postprandial condition), with previous evidence suggesting an interaction between triglyceride-rich lipoproteins and Lp(a) in the postprandial state (39, 40).

Collectively, our findings suggest that the catabolism of Lp(a) does not play a significant role in determining the concentration of Lp(a) particles in patients taking statin therapy across a wide range of Lp(a) concentrations. Our findings that the FCR and PR of apo(a) were positively correlated in patients with elevated Lp(a) concentration may reflect a balancing, feed-forward mechanism in response to increased production of apo(a) in these patients (41). Consistent with this, we previously found similar positive associations between the FCR and PR of very-low-density lipoprotein (VLDL)-apoC-III and HDL-apoA-I in patients with dyslipidemia (42, 43). This phenomenon entails a balancing feedback, which is a feature of complex metabolic systems, as also demonstrated following therapeutic interventions (18, 44). In

contrast, our finding of no significant association between the FCR and PR of apo(a) in patients with normal Lp(a) level suggests the aforementioned balancing feedback mechanism is not operational under normal physiological conditions (16, 18, 44). However, this speculation requires further investigation.

Previous experimental data suggest that insulin may suppress apo(a) synthesis in HepG2 cells (45). However, a prolonged hyperinsulinemic-euglycemic clamp was not shown to alter plasma Lp(a) concentration in healthy individuals (46). Insulin is known to regulate hepatic secretion of VLDL particles in humans (24), but whether this also affects apo(a) production remains unclear. Our finding of no significant association between plasma insulin concentration and the FCR and PR of apo(a) does not support the role of insulin in regulating Lp(a) metabolism in humans. Consistent with this, we recently found no significant association between the FCR of apo(a) and VLDL-apoB-100 ($r = 0.1879$; $P > 0.05$) or between the PR of apo(a) and VLDL-apoB-100 ($r = -0.004$; $P > 0.05$) (19).

Limitations

Our study does have limitations. The sample size was relatively small but of adequate statistical power and larger than sample sizes in previous kinetic studies of Lp(a) metabolism (30, 31, 47). We studied only white subjects to offset potential confounding effects of other ethnicities (48). Hence, our findings may be different in nonwhites. We did not measure the kinetics of apoB-100 in Lp(a) particles. However, we previously demonstrated that the FCRs of apo(a) and apoB-100 were highly correlated in subjects with a wide range of Lp(a) concentrations (18). We measured only the turnover of apo(a) isolated from whole plasma; however, we previously demonstrated that it was equivalent to that of apo(a) isolated from the LDL-HDL fractions (18). Further studies are needed to investigate whether the findings of our study apply to patients with genotypically defined FH, hypertriglyceridemia, and/or diabetes. We studied only patients receiving statins. However, we previously demonstrated that high-dose atorvastatin (80 mg/d) did not significantly alter Lp(a) kinetics in subjects with average Lp(a) concentrations within the normal range (18). Whether these findings apply to subjects with elevated Lp(a) concentrations needs further investigation. Recent experimental data suggest that statin treatment increases LPA expression and apo(a) production in HepG2 cells (14); whether this translates into a clinically meaningful elevation in the hepatic secretion of Lp(a) particles is unknown. We did not measure the kinetics of both apo(a) isoforms because our gel separation method was not sensitive for isolating minor apo(a) isoforms,

especially in patients with low apo(a) concentrations. Hence, the estimation of Lp(a) kinetics based on the predominant isoform may not be accurate, particularly in subjects with two codominant isoforms. Our correlational data between apo(a) isoform size, concentration, and PR might have been different if both apo(a) isoforms were studied. Given that all patients had a predominant apo(a) isoform that contributed $>85\%$ to the total plasma apo(a) concentration, we consider that our kinetic data will reasonably reflect the “true” metabolism of apo(a) in the two groups.

Conclusions

Patients with elevated Lp(a) concentrations remain at high risk of ASCVD despite maximal reduction of LDL-cholesterol with high-intensity statins (7, 8). Our studies of Lp(a) metabolism provide knowledge of the kinetic mechanism for elevated Lp(a) in patients taking optimal cholesterol-lowering therapy. We demonstrated that increased Lp(a) concentration is a consequence of increased production of Lp(a) particles. This supports the clinical use of agents that lower production of Lp(a) (18, 33), with the aim of targeting elevated plasma levels of Lp(a) and attenuating the risk of ASCVD against background statin therapy (17). Future studies of therapies that target the kinetic defect in Lp(a) metabolism that we have demonstrated and their effect on clinical outcomes are required (17).

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Disclosure Summary: The authors declare no conflicts of interest.

Data Availability: The datasets generated during and/or analyzed during the current study are not publicly available but are available from the corresponding author on reasonable request.

References and Notes

- Schmidt K, Noureen A, Kronenberg F, Utermann G. Structure, function, and genetics of lipoprotein (a). *J Lipid Res.* 2016;57(8):1339–1359.
- Reyes-Soffer G, Ginsberg HN, Ramakrishnan R. The metabolism of lipoprotein (a): an ever-evolving story. *J Lipid Res.* 2017;58(9):1756–1764.
- Nordestgaard BG, Chapman MJ, Ray K, Borén J, Andreotti F, Watts GF, Ginsberg H, Amarencu P, Catapano A, Descamps OS, Fisher E, Kovanen PT, Kuivenhoven JA, Lesnik P, Masana L, Reiner Z, Taskinen MR, Tokgözoğlu L, Tybjaerg-Hansen A; European Atherosclerosis Society Consensus Panel. Lipoprotein(a) as a cardiovascular risk factor: current status. *Eur Heart J.* 2010;31(23):2844–2853.
- Danesh J, Collins R, Peto R. Lipoprotein(a) and coronary heart disease: meta-analysis of prospective studies. *Circulation.* 2000;102(10):1082–1085.
- Kamstrup PR, Tybjaerg-Hansen A, Steffensen R, Nordestgaard BG. Genetically elevated lipoprotein(a) and increased risk of myocardial infarction. *JAMA.* 2009;301(22):2331–2339.
- Saleheen D, Haycock PC, Zhao W, Rasheed A, Taleb A, Imran A, Abbas S, Majeed F, Akhtar S, Qamar N, Zaman KS, Yaqoob Z, Saghir T, Rizvi SNH, Memon A, Mallick NH, Ishaq M, Rasheed SZ, Memon FU, Mahmood K, Ahmed N, Frossard P, Tsimikas S, Witztum JL, Marcovina S, Sandhu M, Rader DJ, Danesh J. Apolipoprotein(a) isoform size, lipoprotein(a) concentration, and coronary artery disease: a Mendelian randomisation analysis. *Lancet Diabetes Endocrinol.* 2017;5(7):524–533.
- Khera AV, Everett BM, Caulfield MP, Hantash FM, Wohlgemuth J, Ridker PM, Mora S. Lipoprotein(a) concentrations, rosuvastatin therapy, and residual vascular risk: an analysis from the JUPITER Trial (Justification for the Use of Statins in Prevention: an Intervention Trial Evaluating Rosuvastatin). *Circulation.* 2014;129(6):635–642.
- Nicholls SJ, Tang WH, Scoffone H, Brennan DM, Hartiala J, Allayee H, Hazen SL. Lipoprotein(a) levels and long-term cardiovascular risk in the contemporary era of statin therapy. *J Lipid Res.* 2010;51(10):3055–3061.
- Ooi EMM, Ellis KL, Barrett PHR, Watts GF, Hung J, Beilby JP, Thompson PL, Stobie P, McQuillan BM. Lipoprotein(a) and apolipoprotein(a) isoform size: associations with angiographic extent and severity of coronary artery disease, and carotid artery plaque. *Atherosclerosis.* 2018;275:232–238.
- Havekes L, Vermeer BJ, Brugman T, Emeis J. Binding of LP(a) to the low density lipoprotein receptor of human fibroblasts. *FEBS Lett.* 1981;132(2):169–173.
- Krempler F, Kostner GM, Roscher A, Haslauer F, Bolzano K, Sandhofer F. Studies on the role of specific cell surface receptors in the removal of lipoprotein (a) in man. *J Clin Invest.* 1983;71(5):1431–1441.
- Hofmann SL, Eaton DL, Brown MS, McConathy WJ, Goldstein JL, Hammer RE. Overexpression of human low density lipoprotein receptors leads to accelerated catabolism of Lp(a) lipoprotein in transgenic mice. *J Clin Invest.* 1990;85(5):1542–1547.
- Rader DJ, Mann WA, Cain W, Kraft HG, Usher D, Zech LA, Hoeg JM, Davignon J, Lupien P, Grossman M. The low density lipoprotein receptor is not required for normal catabolism of Lp(a) in humans. *J Clin Invest.* 1995;95(3):1403–1408.
- Tsimikas S, Gortds PLSM, Nora C, Yeang C, Witztum JL. Statin therapy increases lipoprotein(a) levels [published online ahead of print 20 May 2019]. *Eur Heart J.* doi: 10.1093/eurheartj/ehz310.
- Hoover-Plow J, Huang M. Lipoprotein(a) metabolism: potential sites for therapeutic targets. *Metabolism.* 2013;62(4):479–491.
- Barrett PHR, Chan DC, Watts GF. Thematic review series: patient-oriented research: design and analysis of lipoprotein tracer kinetics studies in humans. *J Lipid Res.* 2006;47(8):1607–1619.
- Tsimikas S, Fazio S, Ferdinand KC, Ginsberg HN, Koschinsky ML, Marcovina SM, Moriarty PM, Rader DJ, Remaley AT, Reyes-Soffer G, Santos RD, Thanassoulis G, Witztum JL, Danthi S, Olive M, Liu L. NHLBI Working Group recommendations to reduce lipoprotein(a)-mediated risk of cardiovascular disease and aortic stenosis. *J Am Coll Cardiol.* 2018;71(2):177–192.
- Watts GF, Chan DC, Somaratne R, Wasserman SM, Scott R, Marcovina SM, Barrett PHR. Controlled study of the effect of proprotein convertase subtilisin-kexin type 9 inhibition with evolocumab on lipoprotein(a) particle kinetics. *Eur Heart J.* 2018;39(27):2577–2585.
- Chan DC, Watts GF, Coll B, Wasserman SM, Marcovina SM, Barrett PHR. Lipoprotein(a) particle production as a determinant of plasma lipoprotein(a) concentration across varying apolipoprotein(a) isoform sizes and background cholesterol-lowering therapy. *J Am Heart Assoc.* 2019;8(7):e011781.
- Dwyer KP, Barrett PHR, Chan D, Foo JI, Watts GF, Croft KD. Oxazolinone derivative of leucine for GC-MS: a sensitive and robust method for stable isotope kinetic studies of lipoproteins. *J Lipid Res.* 2002;43(2):344–349.
- Marcovina SM, Albers JJ, Scanu AM, Kennedy H, Giaculli F, Berg K, Couderc R, Dati F, Rifai N, Sakurabayashi I, Tate JR, Steinmetz A. Use of a reference material proposed by the International Federation of Clinical Chemistry and Laboratory Medicine to evaluate analytical methods for the determination of plasma lipoprotein(a). *Clin Chem.* 2000;46(12):1956–1967.
- Marcovina SM, Hobbs HH, Albers JJ. Relation between number of apolipoprotein(a) kringle 4 repeats and mobility of isoforms in agarose gel: basis for a standardized isoform nomenclature. *Clin Chem.* 1996;42(3):436–439.
- Akaike H. A new look at the statistical model identification. *IEEE Trans Automat Contr.* 1974;19(6):716–723.
- Riches FM, Watts GF, Naoumova RP, Kelly JM, Croft KD, Thompson GR. Hepatic secretion of very-low-density lipoprotein apolipoprotein B-100 studied with a stable isotope technique in men with visceral obesity. *Int J Obes Relat Metab Disord.* 1998;22(5):414–423.
- World Health Organization. Familial hypercholesterolemia (FH): report of a second WHO consultation. Geneva, Switzerland: World Health Organization; 1999. Available at: http://apps.who.int/iris/bitstream/10665/66346/1/WHO_HGN_FH_CONS_99.2.pdf. Accessed 2 October 2019.
- Retzlaff JA, Tauxe WN, Kiely JM, Stroebel CF. Erythrocyte volume, plasma volume, and lean body mass in adult men and women. *Blood.* 1969;33(5):649–661.
- Ma L, Chan DC, Ooi EMM, Marcovina SM, Barrett PHR, Watts GF. Data from: Apolipoprotein(a) kinetics in statin-treated patients with elevated plasma lipoprotein(a) concentration. figshare 2019. Deposited 19 June 2019. <http://doi.org/10.6084/m9.figshare.8293250>.
- Rader DJ, Cain W, Ikewaki K, Talley G, Zech LA, Usher D, Brewer HB Jr. The inverse association of plasma lipoprotein(a) concentrations with apolipoprotein(a) isoform size is not due to differences in Lp(a) catabolism but to differences in production rate. *J Clin Invest.* 1994;93(6):2758–2763.
- Rader DJ, Cain W, Zech LA, Usher D, Brewer HB, Jr. Variation in lipoprotein(a) concentrations among individuals with the same apolipoprotein (a) isoform is determined by the rate of lipoprotein(a) production. *J Clin Invest.* 1993;91(2):443–447.
- Frischmann ME, Ikewaki K, Trenkwalder E, Lamina C, Dieplinger B, Soufi M, Schweer H, Schaefer JR, König P, Kronenberg F, Dieplinger H. *In vivo* stable-isotope kinetic study suggests intracellular assembly of lipoprotein(a). *Atherosclerosis.* 2012;225(2):322–327.
- Jenner JL, Seman LJ, Millar JS, Lamon-Fava S, Welty FK, Dolnikowski GG, Marcovina SM, Lichtenstein AH, Barrett PHR, deLuca C, Schaefer EJ. The metabolism of apolipoproteins (a) and B-100 within plasma lipoprotein (a) in human beings. *Metabolism.* 2005;54(3):361–369.
- Reyes-Soffer G, Pavlyha M, Ngai C, Thomas T, Holleran S, Ramakrishnan R, Karmally W, Nandakumar R, Fontanez N, Obunike J, Marcovina SM, Lichtenstein AH, Matthan NR, Matta

- J, Maroccia M, Becue F, Poitiers F, Swanson B, Cowan L, Sasiela WJ, Surks HK, Ginsberg HN. Effects of PCSK9 inhibition with alirocumab on lipoprotein metabolism in healthy humans. *Circulation*. 2017;135(4):352–362.
33. Graham MJ, Viney N, Crooke RM, Tsimikas S. Antisense inhibition of apolipoprotein (a) to lower plasma lipoprotein (a) levels in humans [published correction appears in *J Lipid Res*. 2016; 57(12):2235]. *J Lipid Res*. 2016;57(3):340–351.
 34. Wade DP, Knight BL, Harders-Spengel K, Soutar AK. Detection and quantitation of apolipoprotein(a) mRNA in human liver and its relationship with plasma lipoprotein(a) concentration. *Atherosclerosis*. 1991;91(1-2):63–72.
 35. White AL, Hixson JE, Rainwater DL, Lanford RE. Molecular basis for “null” lipoprotein(a) phenotypes and the influence of apolipoprotein(a) size on plasma lipoprotein(a) level in the baboon. *J Biol Chem*. 1994; 269(12):9060–9066.
 36. Lobentanz E-M, Krasznai K, Gruber A, Brunner C, Müller H-J, Sattler J, Kraft H-G, Utermann G, Dieplinger H. Intracellular metabolism of human apolipoprotein(a) in stably transfected Hep G2 cells. *Biochemistry*. 1998;37(16):5417–5425.
 37. Brunner C, Lobentanz EM, Pethö-Schramm A, Ernst A, Kang C, Dieplinger H, Müller HJ, Utermann G. The number of identical kringle IV repeats in apolipoprotein(a) affects its processing and secretion by HepG2 cells. *J Biol Chem*. 1996; 271(50):32403–32410.
 38. Durovic S, Marz W, Frank S, Scharnagl H, Baumstark MW, Zechner R, Kostner GM. Decreased binding of apo-lipoprotein(a) to familial defective apolipoprotein B-100 (Arg3500Gln): a study of the assembly of recombinant apolipoprotein (a) with mutant low density lipoproteins. *J Biol Chem*. 1994;269(48):30320–30325.
 39. Pfaffinger D, Schuelke J, Kim C, Fless GM, Scanu AM. Relationship between apo[a] isoforms and Lp[a] density in subjects with different apo[a] phenotype: a study before and after a fatty meal. *J Lipid Res*. 1991;32(4):679–683.
 40. Ooi EM, Watts GF, Chan DC, Pang J, Tenneti VS, Hamilton SJ, McCormick SP, Marcovina SM, Barrett PH. Effects of extended-release niacin on the postprandial metabolism of Lp(a) and ApoB-100-containing lipoproteins in statin-treated men with type 2 diabetes mellitus. *Arterioscler Thromb Vasc Biol*. 2015;35(12): 2686–2693.
 41. Csete ME, Doyle JC. Reverse engineering of biological complexity. *Science*. 2002;295(5560):1664–1669.
 42. Chan DC, Nguyen MN, Watts GF, Barrett PH. Plasma apolipoprotein C-III transport in centrally obese men: associations with very low-density lipoprotein apolipoprotein B and high-density lipoprotein apolipoprotein A-I metabolism. *J Clin Endocrinol Metab*. 2008;93(2):557–564.
 43. Chan DC, Barrett PH, Ooi EM, Ji J, Chan DT, Watts GF. Very low density lipoprotein metabolism and plasma adiponectin as predictors of high-density lipoprotein apolipoprotein A-I kinetics in obese and nonobese men. *J Clin Endocrinol Metab*. 2009;94(3): 989–997.
 44. Chan DC, Watts GF, Nguyen MN, Barrett PH. Factorial study of the effect of n-3 fatty acid supplementation and atorvastatin on the kinetics of HDL apolipoproteins A-I and A-II in men with abdominal obesity. *Am J Clin Nutr*. 2006;84(1):37–43.
 45. Riemens SC, Dullaart RP. Effect of insulin on apolipoprotein(a) synthesis. *Diabetologia*. 1999;42(7):A897–A898.
 46. Riemens SC, Dullaart RP. Twenty-four hours of insulin infusion does not lower plasma lipoprotein(a) in healthy men. *Scand J Clin Lab Invest*. 2000;60(2):119–123.
 47. Diffenderfer MR, Lamon-Fava S, Marcovina SM, Barrett PH, Lel J, Dolnikowski GG, Berglund L, Schaefer EJ. Distinct metabolism of apolipoproteins (a) and B-100 within plasma lipoprotein(a). *Metabolism*. 2016;65(4):381–390.
 48. Enkhmaa B, Anuurad E, Berglund L. Lipoprotein (a): impact by ethnicity and environmental and medical conditions. *J Lipid Res*. 2016;57(7):1111–1125.